DriveCompose: Compositional Generative Reconstruction of Driving Scenes via Joint Completion

Anonymous CVPR submission

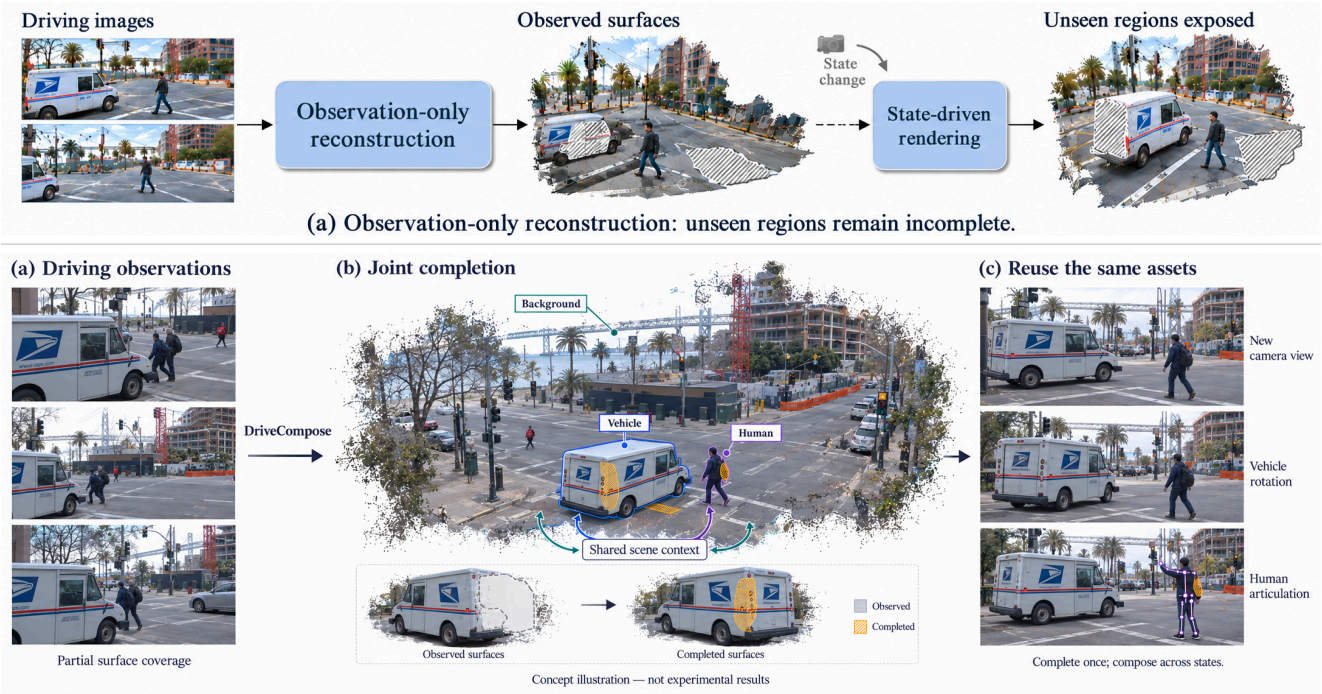

Figure 1. **From observations to reusable completed assets.** Top: viewpoint and actor changes expose missing surfaces in observation-only reconstruction. Bottom: structured encoding and joint completion produce background, canonical vehicles, and humans for reuse across states. Gold indicates completed regions. All scene and asset images are schematic illustrations.

Abstract

Driving simulation and scene editing require reconstructed scenes that support camera and actor motion beyond recorded configurations. Such changes expose unseen surfaces whose geometry and appearance must be completed in the corresponding background, vehicle, or human representation. We present DriveCompose, a framework for compositional generative reconstruction from driving image sequences. Visual features are aggregated in world coordinates for the background and canonical frames for actors, then encoded into component-specific latents. A conditional flow model uses cross-component memory to exchange scene context and jointly complete missing codes. Type-specific Gaussian decoders retain rigid and skeletal bindings, allowing the same completed assets to follow new camera and actor states. Hard latent restoration preserves observed Gaussian parameters, and free-ray constraints control the rendering effects of added content. Under shared states on Waymo, DriveCompose improves reconstruction and off-trajectory rendering over state-adapted baselines; cross-component memory adds 0.627 dB full-image PSNR. Real-scene edits achieve 92.78% success across nine training/generation-seed combinations, demonstrating reuse of fixed completed assets.

1. Introduction

Driving simulation and scene editing require 3D scenes that support changes in camera viewpoint, vehicle motion, and human pose [46, 49]. However, driving images reveal only the surfaces visible from the recorded viewpoints and actor configurations [1, 31]. Moving a camera, turning a vehicle,

or changing a pedestrian's pose exposes previously unobserved regions (Fig. 1). Reconstructing scenes for these operations therefore requires completing the missing geometry and appearance while ensuring that completed vehicle and human surfaces move with their respective actors.

Recent feed-forward methods fuse multiple images to directly predict driving-scene geometry and Gaussian representations [3, 6, 45, 48]. Motion and visibility estimates align dynamic objects and aggregate surfaces visible across the input sequence. When a new viewpoint or actor motion reveals surfaces unseen throughout the sequence, reconstruction must infer their geometry and appearance.

Generative reconstruction uses asset priors, Gaussian completion, or generated views to infer unobserved content [17, 44, 47, 51]. Meanwhile, compositional reconstruction represents the background and moving actors separately to support vehicle motion and human pose control [5, 43, 49]. Applying generative completion to such scenes requires generated surfaces to move with their actors. When an actor is only partially observed, information from the background and other actors can provide additional context for inferring its missing content.

We introduce DriveCompose, which reconstructs driving image sequences into a generatively completed 3D scene with independently controllable actors. We represent the background in world coordinates, rigid surfaces in each asset's own coordinate frame, and human surfaces in a reference pose with skeletal bindings [5, 43]. Completion operates directly on these representations, allowing generated rigid and human surfaces to follow their actors' motion and pose changes (Sec. 3.1). The model jointly completes missing background and actor content using shared scene context.

Figure 2 summarizes construction and reuse. A visual encoder recovers geometry and motion; canonical aggregation aligns observations, and type-specific encoders produce component latents. Local attention models each component, while cross-component memory exchanges scene context to complete missing content through flow matching [18]. Gaussian decoders retain rigid and skeletal bindings [15], allowing state changes to reuse the same completed content (Sec. 3.6).

Under shared camera and actor states on Waymo, DriveCompose improves reconstruction and off-trajectory rendering. Paired synthetic tests show better newly visible surfaces than independent completion; real edits reuse the same completed assets across actor states (Sec. 4).

Our contributions are:

- We introduce DriveCompose to jointly complete backgrounds, vehicles, and humans from image sequences as compositional Gaussian assets, reusable under camera, rigid, and skeletal state changes.
- We develop a cross-component latent completion model based on conditional flow matching. Local attention models spatial relationships within each component, while cross-component memory aggregates background and actor features to provide scene context for generating each component's missing content.
- We evaluate conditional reconstruction, off-trajectory rendering, asset edits, and component ablations. Visibility-stratified diagnostics localize the gains from joint completion to severely missing background and actor regions.

2. Related work

Driving-scene reconstruction. NeRF and Gaussian splatting provide the radiance representations underlying urban reconstruction [15, 22], with driving-specific rendering developed in NeuRAD and SplatAD [11, 33]. Feed-forward methods learn geometry and Gaussian prediction from images [2, 4, 12, 35, 36]. For driving logs, STORM aggregates dynamic Gaussians through scene flow, while DGGT combines camera recovery with temporal visibility and motion-guided interpolation [3, 45]. DrivingForward, StreetForward, PointForward, and Instant NuRec further explore surround-view inputs, temporal processing, point-aligned representations, and simulation-oriented reconstruction [6, 14, 32, 48]. TokenGS also decouples Gaussian prediction from pixels through learnable tokens [29].

Compositional scenes and controllable assets. Neural Scene Graphs, Street Gaussians, and DrivingGaussian separate scene content into background and moving components [23, 43, 50]. OmniRe includes canonical representations for rigid and articulated actors, while UniSim and HUGSIM connect reconstructed content to closed-loop simulation [5, 46, 49]. At the asset level, SMPL-based body models and animatable Gaussian methods provide canonical surfaces and motion bindings [13, 16, 20, 25, 27]; LHM and IDOL infer animatable humans, and UrbanCAD targets controllable vehicles [21, 28, 52]. Our completion variables retain these established component identities, frames, and bindings.

Generative reconstruction and completion. Image-conditioned 3D generation uses generated views [9, 19, 30] or structured latents encoding geometry and appearance [41, 42]. GenAssets learns driving-asset priors, CompleteSplat completes hidden Gaussian content, and JRM aggregates unaligned object observations through latent flow matching [17, 40, 47]. LatentWorld organizes semantic occupancy generation around entity and background latents [24]. DriveCompose targets RGB-renderable reconstruction conditioned on driving logs: a shared completion model exchanges context across world-space background

and canonical rigid and human codes.

Generative driving scenes and videos. MagicDrive, DriveDreamer, and Vista generate driving imagery conditioned on scene structure or control [8, 10, 37]. StreetCrafter uses generated views to supervise dynamic Gaussian reconstruction; HorizonForge combines scene editing with video diffusion [38, 44]. WorldSplat generates Gaussians and refines their rendered videos, while DIFIX3D+ applies diffusion-based image refinement [39, 51]. DriveCompose completes and stores background and canonical assets once, then renders them under new camera and actor states.

3. Method

DriveCompose aligns multi-view observations on canonical supports, jointly completes their latents, and composes the decoded assets under requested states (Fig. 2).

3.1. Problem formulation and compositional representation

The input is a synchronized image log $\mathbf{I}_{t,v}$ with timestamps, camera intrinsics, and camera-to-rig calibration. The controlled evaluations provide rig poses, instance identities, vehicle boxes and root trajectories, and target human joints. Human shape and binding are recovered from context; no target RGB or test LiDAR enters reconstruction. Coordinate conventions and image preprocessing are detailed in Appendix D of the supplementary material.

The output is a compositional Gaussian scene

$$\mathcal{W} = \big(\mathcal{G}_b, C_{\text{sky}}, V_{\text{sky}}, \{\mathcal{A}_a\}, \mathcal{B}, \xi\big), \tag{1}$$

where $\mathcal{G}_b$ is the world-space background, C_{sky} a sky cubemap, V_{sky} its supported directions, $\mathcal{A}_a$ the canonical asset for identity a, $\mathcal{B}$ the finite spatial support, and ξ the sampling seed. The components are rendered by Gaussian splatting [15]. Vehicles are represented as rigid assets with fixed local geometry and dimensions. Humans additionally retain a rest skeleton and skinning weights. Scene state controls the placement and articulation of these fixed components.

We model missing content through conditional generation. Let $\mathcal{Q}$ be the recovered query layout and C its observed evidence. The completion model samples unknown codes $\mathbf{z}_u$, while the codec D_ψ decodes the assembled codes into scene content:

$$\mathbf{z}_u \sim p_\theta(\mathbf{z}_u \mid C, \mathcal{Q}), \qquad \mathcal{W} = D_\psi(\mathbf{z}^{\text{obs}}, \mathbf{z}_u; \mathcal{Q}). \tag{2}$$

The query layout assigns each latent code to a spatial location and scene component. Completion fills the unknown codes within this layout, and decoding preserves their component identities and motion bindings.

3.2. Visual recovery and canonical aggregation

The Driving Log Encoder extracts patch features from each image and exchanges pooled context across cameras and timestamps [7, 34]. Geometry heads predict camera-axis depth, normals, confidence, semantics, and instance assignments. These predictions connect the image features to 3D surfaces and their owning components. Trajectory-level identity slots separate persistent identity, dimensions, and human shape from frame-dependent root poses and joint rotations. Architectural details and state parameterization are given in Appendix D of the supplementary material.

Spatial supports. A query stores canonical position $\mathbf{q}_i$, cell scale $\boldsymbol{\delta}_i$, type k_i, local attention group g_i, and reference-world coordinate $\mathbf{q}_i^w$. Reliable depth samples provide surface anchors. Background anchors are combined with a finite world grid; rigid anchors are aligned by inverse SE(3) transforms and combined with a grid in the local bounding volume. Human queries lie on the canonical template surface. Sky queries are unit directions grouped by cubemap face. These supports provide explicit locations for both observed and missing content.

The fixed layout stores canonical coordinates for component identity and first-input world coordinates $\mathbf{q}_i^w$ (including human skinning) for relating components to their surroundings.

Motion-aware evidence aggregation. For each observation, we place the canonical query at the corresponding actor state:

$$\mathbf{p}_i(t) = \begin{cases} \mathbf{q}_i, & k_i = \text{background}, \\ \text{xyz}(\mathbf{T}_a^w(t)\bar{\mathbf{q}}_i), & k_i = \text{rigid}, \\ \text{xyz}(\mathbf{T}_h^w(t)\mathbf{A}_i(t)\bar{\mathbf{q}}_i), & k_i = \text{human}. \end{cases} \tag{3}$$

Here $\bar{\mathbf{q}}_i$ is homogeneous, xyz extracts Euclidean coordinates, and $\mathbf{A}_i(t)$ is the weighted skinning transform. Projection gives query depth $z_{i,t,v}$ and samples image features $\boldsymbol{\phi}_{i,t,v}$, depth $d_{i,t,v}$, and confidence $c_{i,t,v}$. We weight observations by confidence and normalized depth residual $r = (d - z)/(\eta_{\text{abs}} + \eta_{\text{rel}} d)$. The indicator $a_{i,t,v}$ selects visible, depth-consistent observations of the same instance (Appendix D):

$$w_{i,t,v} = a_{i,t,v} c_{i,t,v} \exp(-r_{i,t,v}^2/2), \tag{4}$$

$$\mathbf{h}_i = \frac{\sum_{t,v} w_{i,t,v}[\boldsymbol{\phi}_{i,t,v}; r_{i,t,v}]}{\max(\varepsilon, \sum_{t,v} w_{i,t,v})}. \tag{5}$$

The masks m_i, f_i, and u_i distinguish observed surfaces, ray-labeled free space, and unknown content, with $m_i + f_i +$

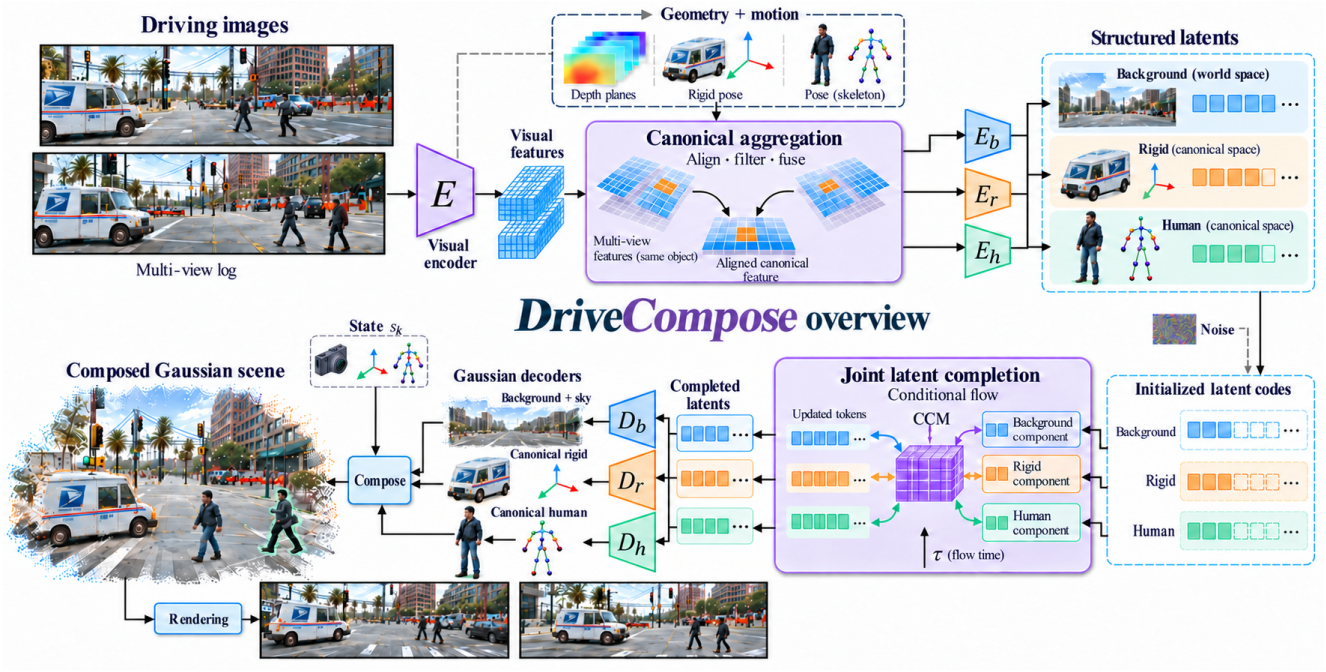

Figure 2. **DriveCompose overview.** Visual encoding and canonical aggregation recover component features; type-specific encoders, joint completion, and Gaussian decoders produce background, vehicle, and human content for state-driven composition. Scene and asset images are schematic illustrations.

$u_i = 1$. Confidence is the maximum over valid observations; two reliable depth-ray votes establish a free-space label. Sky features are aggregated by viewing direction using camera rotation and sky semantics. Appendix D gives the criteria and a footprint-based mask audit.

3.3. Latent representation encoding

Type-specific encoders map the 129-dimensional aggregate and its spatial support, $[\mathbf{h}_i; \mathbf{q}_i; \boldsymbol{\delta}_i]$, through a $135 \to 128 \to 128$ MLP with LayerNorm. The resulting codes have dimension $D = 128$ and retain query type, ownership, and frame. Partial and complete branches share this map; masks and confidence condition completion separately. Unobserved codes are zeroed, and observed codes are denoted $\mathbf{z}^{\text{obs}}$.

3.4. Joint latent-space completion

Generative completion module. Our generative completion module (GCM) combines a scene-conditioned flow transformer with constrained sampling (Fig. 3). We parameterize its completion velocity with a DiT-style transformer [26] trained by flow matching [18]. Its input combines current and observed codes, masks, confidence, type, both coordinate descriptions, cell scales, and pooled log features. Background groups follow spatial neighborhoods, each asset forms a local group, and sky directions are grouped by cubemap face. Local self-attention processes these groups separately. A cross-component memory (CCM) module then exchanges context through shared memory:

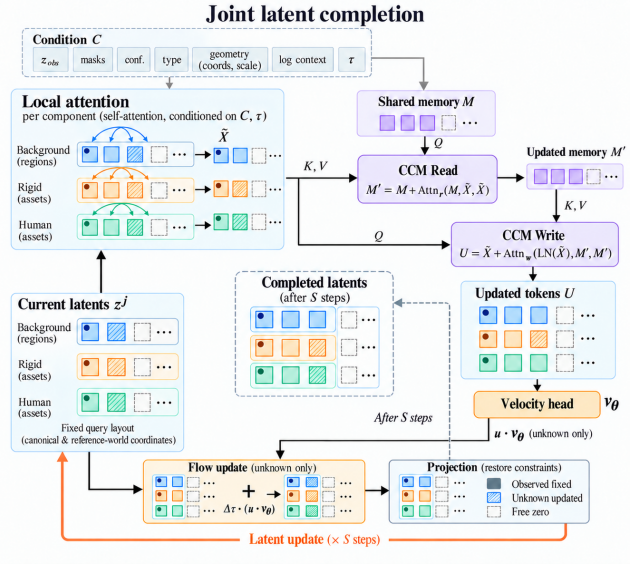

Figure 3. **Joint latent completion.** CCM reads local features and writes back shared context; projected flow updates only unknown codes. Appendix L provides an enlarged diagram.

$$\mathbf{M}' = \mathbf{M} + \mathrm{Attn}_r(\mathbf{M}, \widetilde{\mathbf{X}}, \widetilde{\mathbf{X}}), \tag{6}$$

$$\mathbf{U} = \widetilde{\mathbf{X}} + \mathrm{Attn}_w(\mathrm{LN}(\widetilde{\mathbf{X}}), \mathbf{M}', \mathbf{M}'). \tag{7}$$

Here $\widetilde{\mathbf{X}}$ concatenates local tokens; attention arguments are queries, keys, and values. A residual feed-forward update follows $\mathbf{U}$. Flow time modulates local normalization, and learned tokens plus log summaries initialize memory.

Flow training and constrained sampling. Let $\mathbf{z}_1$ be a complete target on the same query layout and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ be Gaussian noise. We interpolate only the unknown content at flow time $\tau \in [0, 1]$:

$$\mathbf{z}_\tau = m \odot \mathbf{z}^{\mathrm{obs}} + u \odot [(1-\tau)\boldsymbol{\epsilon} + \tau \mathbf{z}_1]. \tag{8}$$

Observed codes remain fixed and free-space codes remain zero. The target velocity for unknown queries is $\mathbf{z}_1 - \boldsymbol{\epsilon}$. With an optional loss selector s_i, the objective is

$$\mathcal{L}_{\mathrm{flow}} = \frac{\sum_i u_i s_i \|\mathbf{v}_\theta(\mathbf{z}_\tau, \tau, C)_i - (\mathbf{z}_{1,i} - \boldsymbol{\epsilon}_i)\|_2^2}{D \max(1, \sum_i u_i s_i)}. \tag{9}$$

Prior learning uses samples with valid targets at every unknown query; s_i selects loss terms within these targets (Sec. 3.7).

At inference, we initialize unknown codes with noise and take S Euler steps with $\Delta\tau = 1/S$:

$$\mathbf{z}^{j+1} = \Pi_{m,f}\left[\mathbf{z}^j + \Delta\tau\, u \odot \mathbf{v}_\theta(\mathbf{z}^j, j/S, C)\right]. \tag{10}$$

After every step, $\Pi_{m,f}$ restores observed codes and clears free-space codes. The seed fixes a completion, decoded once for reuse across states.

3.5. Type-specific Gaussian decoding

Background, rigid, and human Gaussian decoders independently map each query to $K = 4$ child Gaussians whose centers lie in its support cell. Human children inherit the query's skinning weights; rigid children share their asset's rigid transform. Opacity is multiplied by $1 - f_i$, so free queries cannot produce visible Gaussians. A separate sky head predicts cubemap colors within the observation-supported domain V_{sky}. Gaussian and appearance parameterizations are detailed in Appendix E of the supplementary material.

Each query is decoded independently. Freezing the codec and retaining observed codes therefore preserves observed Gaussian parameters. Free-ray losses constrain the visibility and alpha-compositing effects of added Gaussians, measured in Sec. 4.5.

3.6. Scene composition

State-driven rendering. At time step k, state s_k specifies the ego pose, active assets, rigid transforms, and human root and joint poses. Rigid poses transform canonical Gaussian centers and covariances rigidly. Human joint poses deform them through the stored skinning weights, so completed surfaces move with the corresponding body parts. Appendix C gives the transformations. The renderer composites the background, active assets, and valid sky:

$$o_k = \mathcal{R}(\mathrm{Compose}(\mathcal{W}, s_k), \mathcal{C}_k). \tag{11}$$

The background and canonical assets remain fixed; new camera and actor states reuse the same completed content.

3.7. Training procedure

We train visual recovery, fit the codec, learn the completion prior, and adapt the GCM through rendering. Partial input fixes $\mathcal{Q}$, including cell scales, component ownership, and human bindings. Complete meshes supply surface samples within each cell. Multiview rendering and the frozen visual encoder produce $\mathbf{h}_i^{\mathrm{full}}$, which the shared type-specific encoder maps to $\mathbf{z}_{1,i} = E_{k_i}([\mathbf{h}_i^{\mathrm{full}}; \mathbf{q}_i; \boldsymbol{\delta}_i])$. Surface-free cells receive zero features and opacity supervision.

After codec fitting, recovery and the codec remain frozen. Stage 3 learns from complete synthetic targets at every unknown query. Stage 4 updates only the GCM, mixing complete synthetic samples and the 740 real training logs at 1:3. The four stages use 287,360 updates in total; Appendix B gives target construction, network sizes, and training settings.

4. Experiments

We evaluate conditional reconstruction on Waymo, off-trajectory rendering, and reuse of completed assets. Component, visibility, and observation-preservation tests examine where joint completion helps. Appendix A reports zero-shot transfer to nuScenes.

4.1. Experimental setup

Data and supervision. We divide Waymo's 798 training sequences into 740 training and 58 development sequences, evaluating on 202 validation sequences. Synthetic training uses 10,000 scenes, 80 maps, 10,000 vehicles, and 2,500 human identities. Source maps, vehicle models, and human identities are split before creating appearance/motion variants; real logs are separated by road segment, capture, and identifiable actors. Appendix B gives all split sizes and supervision.

Controlled comparison. STORM, DGGT, and StreetForward [3, 45, 48] are retrained from random initialization on the same real logs, synthetic scenes, and complete assets.

Each uses its native supervision; Ours also uses cell-empty and template-binding targets. A shared state adapter assigns ownership and canonicalizes predicted Gaussians for rigid and skeletal placement. Methods share context RGB, calibration, ego poses, vehicle boxes/root tracks, and target human joints; shape and binding come from context. Test-time weights are fixed. Appendix F reports training costs and adaptation details.

Sampling and metrics. Each test sequence supplies 200 frames, forming 2,020 disjoint 20-frame clips at 10 Hz. Three front cameras provide context at $\{0, 5, 10, 15\}$ [45]; the remaining 12 interpolation and four extrapolation frames give 96,960 targets. Extrapolation renders supplied states. We evaluate 240 $\times$ 160 images [14] using full/dynamic PSNR and SSIM, and depth AbsRel/δ_1 at valid LiDAR points. Depth is aligned per frame for evaluation; Appendix F also reports unaligned results. Frames, clips, then sequences are averaged with equal sequence weight.

4.2. Dynamic scene reconstruction

Table 1 aggregates interpolation and extrapolation at 3:1. DriveCompose leads all six metrics, improving full-image and dynamic-region PSNR over the strongest adapted baseline by 1.154 and 2.340 dB. Its full-image PSNR is 28.900 dB for interpolation and 27.556 dB for extrapolation; dynamic-region PSNR is 25.500 and 23.700 dB. Figure 4 shows clearer actor silhouettes and street structure.

Table 1. **Conditional reconstruction on Waymo.** Interpolation and extrapolation are aggregated at 3:1; Appendix F reports both groups.

	Full image		Dynamic Region		Scale-aligned depth	
Method	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	AbsRel ↓	δ_1 ↑
STORM	26.380	0.826	22.100	0.624	0.178	0.741
DGGT	27.410	0.794	22.710	0.651	0.249	0.668
StreetForward	27.010	0.818	20.919	0.592	0.197	0.759
Ours	**28.564**	**0.901**	**25.050**	**0.862**	**0.095**	**0.877**

4.3. Off-trajectory rendering and missing surfaces

We shift cameras laterally by ±1, ±2, and ±4 m while fixing orientation, intrinsics, and timestamps [6, 51]. Gaussian renders are compared before diffusion refinement. Each direction contains 121,200 images and 6,060 20-frame videos at 10 Hz, referenced to recordings from the same clips, cameras, and timestamps. Left/right FID and FVD are computed separately, then averaged (Table 2; Appendix H).

Figure 5 compares shifted renders against an original-trajectory reference. For paired evaluation, 100 synthetic test scenes provide regions unseen in every input. At ±2 m, Ours improves hidden-region PSNR from 21.56 to 23.18 dB over w/o CCM, reduces LPIPS from 0.236 to 0.192, and raises surface coverage from 89.7% to 95.0% and F-score@2 cm from 81.3% to 88.6%. On real scenes, Ours reduces common-support RGB reprojection MAE from 0.028 to 0.019 versus w/o CCM and raises coverage from 86.9% to 92.4%. Appendix H defines surface extraction and correspondence evaluation.

Table 2. **Off-trajectory distribution quality.** FID/FVD are computed separately for left and right shifts, then averaged.

Method	Shift ±1 m		Shift ±2 m		Shift ±4 m	
	FID ↓	FVD ↓	FID ↓	FVD ↓	FID ↓	FVD ↓
STORM	61.97	722.27	83.72	1029.36	132.57	1704.16
DGGT	73.99	984.00	93.36	1197.03	128.01	1661.70
StreetForward	84.30	878.95	103.31	1095.02	123.58	1578.63
Ours	**28.52**	**544.27**	**37.43**	**673.17**	**58.19**	**817.13**

4.4. Ablations and component context

Protocol. Ablations use one 29-timestamp window per test sequence, three cameras, nested inputs of $T = 2, 4, 8$ timestamps, and seven shared targets (Appendix G). Variants share recovery, codec, states, data, and optimization budgets. Training samples T uniformly; inference uses 24 Euler steps. Each of three training seeds is evaluated with three generation seeds, all averaged without best-sample selection. All 10,908 learned and 202 deterministic window instances are renderable; poor-quality outputs remain scored.

Completion and context. The w/o GCM control disables non-observed, non-sky Gaussians. The w/o CCM control replaces shared memory with isolated per-component memories, retaining the same parameters and global log context, and is retrained through Stages 3–4. Table 3 shows full-image/dynamic PSNR gains of 1.776/2.186 dB over w/o GCM and 0.627/0.934 dB over w/o CCM. Full-image PSNR training-seed SDs are 0.061 and 0.074 dB for Ours and w/o CCM. At $T = 2, 4, 8$, fixed-checkpoint PSNR gains are 1.104, 0.738, and 0.627 dB; the corresponding LPIPS reductions are 0.041, 0.028, and 0.018 (Appendix G).

Where CCM helps. On the 100 synthetic scenes, input-union surface coverage defines well-observed ($\geq$ 80%) and severely missing ($\leq$ 20%) groups. CCM adds 1.02/1.28/1.54 dB for severely missing background/vehicle/human regions, versus 0.04/0.04/0.03 dB for well-observed regions. On Waymo, inference-time read masks on the same Ours checkpoints yield

Figure 4. **Conditional dynamic reconstruction.** Rows show five examples; columns are ground truth, DGGT, STORM, StreetForward, and Ours.

Table 3. **Component ablations.** $T = 8$, with shared recovery, codec, and target views. Means cover three training and three generation seeds; full SDs are in Appendix G.

Variant	Full image			Dynamic regions	
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑
w/o GCM	27.137	0.867	0.183	23.431	0.811
w/o CCM	28.286	0.893	0.147	24.683	0.849
Ours	**28.913**	**0.911**	**0.129**	**25.617**	**0.873**

28.721/28.574/28.352 dB for background/other vehicles/other humans, versus 28.913 with all components; the retrained w/o CCM scores 28.286. Appendix H details the read-scope interventions.

4.5. State-driven reuse and observation preservation

Asset edits. Twenty real and twenty identity-disjoint synthetic scenes each contain one tested vehicle and human. A 1 m camera shift, 30° vehicle rotation, and human pose change reuse fixed completed assets. Table 4 evaluates newly visible synthetic surfaces at 2 cm vehicle and 1 cm human coverage thresholds.

Across nine training/generation-seed combinations, real edits succeed for 177/180 vehicles and 157/180 humans, yielding 92.78% overall success. A scene-cluster bootstrap gives a 95% interval of 85.28–99.17%. Vehicle dimensions and human shape remain fixed; previously observed human texture has canonical linear-RGB MAE 0.014. Appendix I reports individual seeds, blinded failure ratings, and elbow, self-occlusion, and clothing failures.

Observation preservation. Compared with the observed-only scene at fixed cameras and actor states, Ours pre-

Figure 5. **Off-trajectory rendering with a 2 m leftward shift.** Columns show GT, STORM, DGGT, StreetForward, and Ours. GT is the original-trajectory image.

Table 4. **Newly visible asset surfaces after editing.** Paired synthetic tests. Coverage uses 2 cm for vehicles and 1 cm for humans; higher PSNR/coverage and lower LPIPS are better.

Asset	Variant	PSNR	LPIPS	Coverage
Vehicle	w/o GCM	17.80	0.362	63.4%
	w/o CCM	22.46	0.218	91.8%
	Ours	**24.18**	**0.176**	**96.1%**
Human	w/o GCM	17.12	0.391	60.8%
	w/o CCM	21.38	0.247	88.6%
	Ours	**23.06**	**0.201**	**94.3%**

serves observed Gaussian parameters exactly, while input-view PSNR decreases by 0.029 dB and depth MAE increases by 2 mm. On 2,020,000 fixed rays, added-occlusion and strict free-ray violation rates are 0.18% and 0.07%; removing the free-ray loss raises them to 1.34% and 0.91%. Appendix J reports parameter and rendering measurements.

Efficiency. On an RTX 4090 (FP16, batch 1), construction takes 2.66 ± 0.11 s and single-camera state composition/rendering takes 7.8 ± 0.4 ms at 240×160, with 14.2 GB peak memory. CCM adds 5.1% construction time. The 1,000-query amortized cost is 10.46 ms per image. Appendix K specifies scene size, synchronization, warmups, and repetitions.

5. Discussion

DriveCompose reconstructs finite regions under fixed illumination, using rigid vehicles and humans bound to a fixed skeleton and clothing volume. Depth, identity, and canonical-pose accuracy determine the observed anchors inherited by completion. Residual reconstruction artifacts occur near pedestrians and the parked pickup in Fig. 4. Editing failures concentrate on large elbow bends, leg self-occlusion, and loose-clothing boundaries. Relighting and clothing-geometry changes call for richer appearance and deformation models.

6. Conclusion

DriveCompose jointly completes world-space background and canonical actor latents, preserving component identity and motion bindings for state-driven reuse. Shared scene context improves missing-surface reconstruction, while fixed Gaussian assets support camera shifts, vehicle rotation, and human pose edits. Waymo reconstruction, paired hidden-surface tests, and real-scene edits demonstrate the quality and reuse of the completed content.

LLM Usage Statement

OpenAI Codex assisted manuscript revision, translation, LaTeX editing, and consistency checks. OpenAI Image Gen was used to produce the conceptual scene and method illustrations in Figs. 1–3 and the supplementary enlarged diagram. These illustrations are identified as schematic. The authors supplied the reported numerical results and retain responsibility for the scientific content.

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